Sandwiches Party

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

You want to make as many sandwiches as possible.

Each sandwich requires	**You currently have**
2 slices of bread	B slices of bread
5 grams of cheese	X grams of cheese
20 grams of meat	Y grams of meat
10 grams of vegetables	Z grams of vegetables
7 milliliters of sauce	M milliliters of sauce

If you don't have enough of any ingredient, you can buy more. You have K coins, and the **prices** of the ingredients as follows:

Ingredient	**Price per unit**
Bread slice	A
Cheese (per gram)	C
Meat (per gram)	D
Vegetables (per gram)	E
Sauce (per milliliter)	F

You may buy any amount of ingredients as long as the **total cost does not exceed K.**

Your task is to determine the **maximum number of sandwiches** you can prepare.

Input

The first line contains five integers B, X, Y, Z, and M ($1 \leq B, X, Y, Z, M \leq 1000$) the amounts of bread slices, cheese (grams), meat (grams), vegetables (grams), and sauce (milliliters) you currently have.

The second line contains five integers A, C, D, E, and F ($1 \leq A, C, D, E, F \leq 1000$) the prices of one bread slice, one gram of cheese, one gram of meat, one gram of vegetables, and one milliliter of sauce.

The third line contains one integer K ($0 \leq K \leq 100$) the amount of money you have.

Output

Print a single integer — the **maximum number** of sandwiches you can prepare.

Examples

standard input	standard output

`30 20 10 10 13 20`
`24 12 12 11 10 9`
`43`
	`0`

`95 792 201 453 290`
`69 17 49 13 41`
`88`
	`10`